



High rates of Parkinson's disease diagnosis in the autistic population: True co-occurrence or a product of overlapping traits?

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ARTICLE INFO

Key words:

Autism
Parkinson's Disease
Movement
Diagnosis

ABSTRACT

Older autistic adult literature is sparse, and little is known about the aging autistic population. However, recent evidence suggests an increased prevalence of Parkinson's disease (PD) diagnosis in the autistic population. It may initially be assumed that autistic individuals are genetically more likely to develop PD, but extant genetic studies do not provide strong evidence for a link between the two conditions. An underappreciated body of evidence may shed light upon why autistic individuals score highly on PD diagnostic criteria: movement differences in autism have been likened to PD. Given that PD diagnosis is primarily movement-based, if it is the case that autistic movement appears parkinsonian, this may facilitate autistic individuals meeting diagnostic criteria for PD. If validated, this theory could have serious implications for the specificity of the PD diagnostic process. Here, we set out the evidence for high rates of PD diagnosis and parkinsonism in the autistic population and subsequently question why this might be the case, making reference to genetic and behavioural similarities between autism and PD.

1. Introduction

Autism is a condition characterised by socio-communicative differences and restricted or repetitive behaviours (American Psychiatric Association, 2013, Keating et al., 2022). However, recent studies have highlighted movement differences in autism (Fournier et al., 2010, Mari et al., 2003, Gowen and Hamilton, 2013, Leary and Hill, 1996). Indeed, recent calls have been made to include movement differences as “specifiers” to autism diagnoses (i.e., condition-relevant but non-selective symptoms used to further clarify a diagnosis; Licari et al., 2022). Parkinson's disease (PD), by contrast, is a neurodegenerative disorder in which movement differences are the primary focus, including bradykinesia (slowed movement), gait differences and postural instability (Zanardi et al., 2021, Jankovic, 2008, Moustafa et al., 2016). Given that a growing body of research has also highlighted differences in social and emotional processing in PD (Czernecki et al., 2021), it is apparent that the two conditions, whilst outwardly appearing vastly different, are both characterised by co-occurring differences in motor function and social cognition (Eddy and Cook, 2018).

Overlapping traits in autism and PD have garnered recent attention, with a number of review articles commenting on similarities at genetic,

clinical and cognitive levels (Hollander et al., 2009, Mai et al., 2023, Morato Torres et al., 2020). Such attention follows reports of alarmingly high rates of PD diagnosis in the autistic population (Hand et al., 2020, Croen et al., 2015). Relatedly, assessments of parkinsonism (a clinical syndrome defined by impaired motor function which includes PD) also appear elevated in autistic individuals compared to non-autistic individuals (Geurts et al., 2022, Starkstein et al., 2015, Mostert-Kerckhoffs et al., 2020, Wallace et al., 2025). However, a key question concerns whether this is because autistic people are predisposed to PD at a biological level, and/or whether older autistic adults exhibit behaviours that look similar to PD without clear biological markers of the condition. Genetic investigations of autism and PD do link a number of PD-specific genes to autism (Yin et al., 2016, Labonne et al., 2020), however a strong genetic overlap between the two conditions is not apparent with existing approaches (Smeland et al., 2021, Sey et al., 2020). Whilst a genetic predisposition to PD in autism is not yet evidenced, a further possibility is that behavioural similarities between autism and PD may contribute to the increased PD diagnosis prevalence in the autistic population. PD assessments are primarily behavioural, and ancillary investigations (such as brain scans) are not common (Bloem et al., 2021). Furthermore, PD diagnosis is heavily reliant on behavioural observations wherein

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<https://doi.org/10.1016/j.neubiorev.2025.106261>

Received 7 February 2025; Received in revised form 7 May 2025; Accepted 13 June 2025

Available online 13 June 2025

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clinicians look for particular body movement and motor control profiles. Given that body movements tend to be different in autism (compared to the non-autistic population), it is possible that PD diagnosis rates in autism may be higher than in the non-autistic population due to movement similarities between autism and PD. Further investigation of this theory becomes increasingly relevant as recent studies have proposed algorithms for detecting PD which rely on movement features (Lamba et al., 2021). In this case, similarities between autistic and parkinsonian movement could lead to autistic individuals being falsely classified into a PD group.

Although early observational studies anecdotally highlight behavioural similarities between autism and parkinsonism (Minshew et al., 2004, Mari et al., 2003, Vilensky et al., 1981), at present, no direct comparison studies exist. Reflecting on the separate autism and PD literature, both conditions appear to show gait differences (Lum et al., 2021, Jankovic, 2008), postural instability (Jankovic, 2008, Minshew et al., 2004) and bradykinesia (Jankovic, 2008, Mari et al., 2003), all core features of PD. In addition, fine motor control similarities have been noted in activities including handwriting (Godde et al., 2018, Van Gemmert et al., 2003), and emotion production (Ricciardi et al., 2015, Trevisan et al., 2018, Keating et al., 2023). Beyond movement, various cognitive similarities are apparent between autism and PD, most notably in domains classically linked to autism including emotion perception (Gray and Tickle-Degnen, 2010, Lozier et al., 2014, Uljarevic and Hamilton, 2013), theory of mind (White et al., 2011, Peron et al., 2009) and cognitive rigidity (Dirnberger and Jahanshahi, 2013, Craig et al., 2016, Westwood et al., 2016). This Mini-Review posits the possibility that such similarities in traits, particularly those relating to motor function, may lead to autistic individuals scoring highly on PD diagnostic criteria even in the absence of PD pathology. However, it must be noted that similarities between conditions are drawn from separate research studies, meaning that results may be confounded by different sample demographics or tasks used. Therefore, future work is needed to confirm similarities between behavioural presentations of autism and PD.

Understanding the nature of the relationship between autism and PD is of great importance for the autistic population. Indeed, participatory research activities with autistic individuals confirm that this group want to see research into aging in autism, co-occurring medical conditions and clearer diagnoses (Pellicano et al., 2014); these topics are directly relevant to the question of why older autistic individuals are more likely to be diagnosed with PD. If it is the case that autistic individuals are biologically predisposed to develop PD this would highlight a need to adapt current clinical procedures. For example, greater awareness of the co-occurrence between the two conditions would be needed, as well as the establishment of interdisciplinary clinical care teams to ensure communication between movement disorder specialists and psychologists (Morato Torres et al., 2020). Conversely, if the overlap between autism and PD is limited to behavioural and cognitive levels, it may be considered that such similarities between the two conditions play a role in the increased prevalence of PD diagnosis in the autistic population. This would have serious implications from a clinical diagnostic point-of-view. If it is the case that older autistic individuals are sometimes falsely diagnosed with PD on the basis of outward appearances, and are potentially receiving unnecessary medications as a result, the PD diagnosis procedure must be overhauled to guard against this.

In this Mini-Review, we present evidence for elevated PD prevalence rates and parkinsonism in the autistic population. Subsequently, we bring together extant literature to claim that, whilst evidence of a genetic overlap between autism and PD is inconclusive with existing approaches, anecdotal evidence for behavioural and cognitive similarities between the two conditions exists. Recommendations for future research are made to enable a greater understanding of why autistic individuals are more likely to receive a PD diagnosis than the non-autistic population.

2. Are Parkinson's disease diagnosis and parkinsonism prevalent in autism?

Shockingly high rates of PD diagnosis are apparent in the autistic population. Croen et al. (2015) reported a PD prevalence of 0.93 % in a sample of autistic adults ($N = 1507$, age mean[standard deviation(SD)] = 29.0[12.2]), compared to 0.03 % in the non-autistic sample adults ($N = 15,070$, age mean[SD] = 29.4[12.2]). Hand et al. (2020) identified a much higher prevalence rate of 6.6 % in a sample of autistic adults over the age of 65 ($N = 4685$), compared to just 1.2 % in the non-autistic population ($N = 46,850$). These studies provide clear evidence that PD diagnoses are much more prevalent in the autistic population than in the non-autistic population.

Moving beyond official diagnosis rates, studies have indicated that autistic individuals score highly on parkinsonism assessments. Starkstein et al. (2015) conducted an examination of parkinsonian motor signs in autistic adults using the Movement Disorders Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS), a measure frequently used to quantify PD traits (Martínez-Martín et al., 1994). They reported that 32 % of their test sample ($N = 37$, age mean[SD] = 51.2[8.5]) met the diagnostic criteria for parkinsonism, with 25 % of individuals not taking antipsychotic medications meeting the criteria. Similarly, a study of autistic adolescents revealed significantly more bradykinetic and rigid motor behaviour in the autistic group (child sample: $N = 22$, age mean[SD] = 10.4[1.7]; adolescent sample: $N = 23$, age mean[SD] = 18.7[4.6]) compared to the non-autistic group (child sample: $N = 22$, age mean[SD] = 10.3[1.8]; adolescent sample: $N = 26$, age mean[SD] = 20.2[4.1] as measured by the UPDRS (Mostert-Kerckhoffs et al., 2020). A further study of 296 autistic participants from the Netherlands (age mean[SD] = 58.4[5.9]) and 209 autistic participants from the USA (age mean[SD] = 59.4[7.2]) demonstrated positive screening for parkinsonism via the Parkinsonism Screening Questionnaire in 17 % and 33 % of autistic participants respectively (Geurts et al., 2022). Finally, Wallace et al. (2025) extended Geurts' USA sample by incorporating additional autistic adults in the age range 40–49 ($N = 379$, age mean[SD] = 51.9[9.1]); 31 % screened positive for parkinsonism and these individuals reported significantly lower quality of life and sleep quality, and significantly higher memory problems and depression symptoms. Importantly, these studies consider the relevance of antipsychotics - medications commonly prescribed in the autistic population (Houghton et al., 2017) which have been found to cause parkinsonism (Brigo et al., 2014). Both Starkstein and Mostert-Kerckhoff's studies demonstrated elevated parkinsonism in samples of autistic individuals *not* taking antipsychotics. In addition, Geurts' study did not find a difference in antipsychotic use between samples with and without parkinsonism. The relationships between self-reported parkinsonism and other outcomes reported by Wallace and colleagues remained significant after accounting for antipsychotic use. Together these studies demonstrate high levels of parkinsonism in the autistic population, a pattern which cannot be accounted for by antipsychotic use.

3. Are autism and Parkinson's disease genetically related?

Following reports of an increased prevalence of PD diagnosis and parkinsonism in the autistic population (Starkstein et al., 2015, Hand et al., 2020, Croen et al., 2015, Geurts et al., 2022, Mostert-Kerckhoffs et al., 2020, Wallace et al., 2025), it is important to determine why this is the case. It may be that autistic individuals are biologically more likely to develop PD than non-autistic individuals. A crucial step towards answering this question is to determine the genetic relationship between autism and PD – that is, whether there is overlap between the genes associated with autism and the genes associated with PD. Here, we focus on condition-level overlap between autism and PD, given that PD is the condition assessed in the aforementioned prevalence studies and the most common form of parkinsonism. Despite vastly different ages of onset, a lack of temporal overlap in symptom expression does not

preclude the possibility of shared genetic architecture. Indeed, genetic links between neurodevelopmental and neurodegenerative conditions are plausible due to temporal differences in gene expression (modulated by environmental or age-related factors), convergence on core biological pathways, and the potential effects of neurodegeneration-related genes on early cognitive and brain development (Ogonowski et al., 2024).

3.1. Specific genes

Genetic investigations have resulted in the discovery of a number of important risk genes for PD (Nuytemans et al., 2010, Bloem et al., 2021), whereby people with particular variants of these genes have a higher likelihood of obtaining a PD diagnosis in later life. Specific risk genes are associated with variations in age of onset for PD (Lill et al., 2015). Interestingly, many of these genes have also been associated with autism, including PARK2 (Yin et al., 2016, Glessner et al., 2009, Conceição et al., 2017), PINK1 (Zhou et al., 2019) and LRRK2 (Labonne et al., 2020). This suggests that some genetic risk factors for PD may be present in autistic individuals. However, this evidence overlooks the fact that genetic aetiologies for autism and PD are not comprised of a single gene, rather a combination of many risk variants. A broader picture of polygenic overlap is necessary to more fully understand the genetic relationship between autism and PD.

3.2. Genetic overlap

Genome wide association studies (GWAS) have been conducted for both autism and PD in which summary statistics provide a measure of effect size for each single nucleotide polymorphism (SNPs; the most common form of genetic variation) with regards to their association with the condition (Grove et al., 2019, Nalls et al., 2019). These summary statistics can be compared to identify the extent to which SNPs are associated with *both* conditions, for example by employing techniques such as cross-trait Linkage Disequilibrium Score Regression (Bulik-Sullivan et al., 2015). At present, such approaches have yielded null results. The Brainstorm Consortium et al., (2018) reported a non-significant (but numerically negative) genetic correlation between autism and PD (correlation = -0.20) using autism GWAS data from The Autism Spectrum Disorders Working Group Of The Psychiatric Genomics Consortium (2017) and PD GWAS data from Nalls et al. (2019). Sey et al. (2020) also found a non-significant (but numerically positive) genetic correlation between autism and PD (correlation = 0.02) using more recent GWAS data from Grove and colleagues and Nalls and colleagues respectively, and no genetic relationship in an analysis where SNPs were grouped at the gene-level. Finally, Smeland et al. (2021) reported no evidence for overlap in SNP associations for autism and PD when using GWAS data from Grove and colleagues and Nalls and colleagues. Whilst it is possible that relationships may emerge with the development of larger and thus more reliable GWAS, current literature suggests that the genetic bases of autism and PD are not significantly correlated.

Genetic overlap between conditions can also be assessed through the calculation of polygenic risk scores (PRS) - a single value estimate of an individual's propensity to a phenotype, calculated using individual-level genetic data and GWAS data for the phenotype of interest (Choi et al., 2018). Ellis et al. (2020) calculated a range of PRS for 5160 individuals with Huntington's Disease, including those relating to autism and PD using Grove and colleagues and Nalls and colleagues GWAS datasets. No significant correlation was found between autism and PD PRS. Future studies could investigate whether PD PRS are elevated in autistic individuals, specifically those exhibiting movement differences.

3.3. Family studies

A small amount of genetic overlap is expected between first-degree relatives. Therefore, if autistic individuals have increased numbers of

PD risk genes, first-degree relatives of autistic individuals are more likely to have these genes than members of the general population. This means that if autistic individuals are genetically more likely to develop PD, this should extend to first-degree relatives of autistic individuals, albeit to a lesser extent. Interestingly, a study of older individuals with ($N = 739$) and without ($N = 11,666$) a first-degree autistic relative revealed that the two groups did not significantly differ in their rates of PD diagnosis (Stewart et al., 2020). Thus, these results also fail to provide evidence in support of the theory that autistic individuals have a genetic predisposition to PD.

3.4. Biological mechanisms involved in autism and Parkinson's disease

The above sections have focused on genetic risk variants for autism and PD. However, an overlap in functionally linked genes may explain similarities between the two conditions. Functionally linked genes share a common function or are involved in a shared biological pathway. Notably, both autism and PD show associations with genes involved in mitochondrial function (Billingsley et al., 2019a, Castora, 2019, Pecorelli et al., 2020) and synaptic processes (Morato Torres et al., 2020). Thus, genetic convergence may occur at the level of shared biological functions rather than direct risk for either condition. Whilst no single pathway has been robustly implicated in both disorders, future work using broader functional and network-based approaches may implicate overlapping systems in the genetic architecture of autism and PD.

It is additionally plausible that similar behavioural presentations in autism and PD arise from shared neurobiological mechanisms. To date, little work has directly investigated shared biological correlates of autism and PD. An intriguing place to start would be to examine the involvement of the dopamine system in both conditions. Dopamine system dysfunction, specifically degeneration of dopamine neurons in the substantia nigra, is the hallmark of PD (Rizek et al., 2016). Whilst less is known about the pathology of autism, the dopamine hypothesis states that many of the behavioural and cognitive differences observed in autism, including movement, reward processing and repetitive behaviours, may be the result of changes to the dopamine system (Pavál, 2017). Evidence from pharmacology, animal models and PET neuro-imaging leads to conflicting conclusions with respect to a hyper- or hypo-dopaminergic state (Kosillo and Bateup, 2021, Ernst et al., 1997, Nakamura et al., 2010, Hirsch and Pringsheim, 2016, Sharma and Shaw, 2012, Gunaydin et al., 2014, Presti et al., 2003). To the best of our knowledge no studies have directly investigated whether autism and PD share a common biological correlate with respect to dopamine system function (e.g., via DAT-SPECT and PET), making this an important area for future research.

4. Does autistic movement appear parkinsonian?

Moving on from genetic overlap, it should be considered whether elevated PD diagnosis rates in the autistic population stem from the presence of PD-like behaviours – such as movement differences – in the autistic population. In other words, even if autistic people are neither genetically nor biologically predisposed to PD, they may nevertheless receive a PD diagnosis because they exhibit behaviours that appear parkinsonian. Whilst anecdotal evidence supports such similarities (Fig. 1), autistic and parkinsonian movements have never been directly compared. Without such direct comparisons one must rely on interpretation of each separate research silo to build a picture of whether movement similarities exist.

4.1. Gross motor function

Despite a lack of direct comparison studies between autism and PD, there are many early studies of autistic behaviour noting similarities to parkinsonian movement. Bradykinesia – or slowness of movement – is a core feature of PD (Jankovic, 2008) that has also been reported in the

“The connection you are investigating between PD and autism is of interest to me as 18 months prior to my diagnosis my husband used to observe that I didn't swing my arms when walking. He used to liken me to a character called Jake in a drama called Touch. Jake was, I believe, on the autism spectrum.”

“What caught my eyes when I read about the trial on similarities between PD movements and Autism movements is that I have been showing signals of PD since my childhood. On top of that, my sister, who is a doctor, insists that she can see Autism in my behaviour, although I am 56 and I was never diagnosed by a specialist as being autistic.”

“My wife worked with teenagers with Asperger's for many years. Only a couple days ago she said I have started walking in a similar way to some of her students.”

Fig. 1. Lived-experience insights into *similarities between autism and Parkinson's Disease*. Note. Quotes from research participants with Parkinson's Disease referencing observed links to autistic traits.

autistic population (Maurer and Damasio, 1982, Mari et al., 2003). Mari et al. (2003) stated that the slowed movement they observed in autistic children had “a strong resemblance to Parkinsonian-type bradykinesia”. Similarly, individuals with PD often exhibit a complete freezing during their movements, a form of akinesia (Perez-Lloret et al., 2014). This can be likened to catatonia, characterised by a lack of movement, which has been observed in autism (Realmuto and August, 1991, Wing and Shah, 2000, Kakooza-Mwesige et al., 2008, Mazzone et al., 2014). Severe catatonic features have been reported by 17 % of caregivers of autistic individuals over the age of 15 (Wing and Shah, 2000). Thus, a slowness (or absence) of movement is characteristic of both autism and PD.

Gait differences are also apparent in both conditions. Indeed, Vilensky et al. (1981) conducted a study of autistic movement and stated that “the gait differences between the autistic and normal subjects resembled differences between the gaits of parkinsonian patients and of normal adults”. Differences in gait are a core feature of PD (Jankovic, 2008, Mazzoni et al., 2012). Extant literature and subsequent meta-analyses of studies of gait in PD (Zanardi et al., 2021) and autism (Lum et al., 2021) reveal a number of comparable metrics obtained through gait analysis techniques including slower walking speed, reduced stride time, longer double support time and increased stride length variability (Rinehart et al., 2006b, Blin et al., 1990). Thus, as initially highlighted by Vilensky et al. (1981), gait differences are apparent in both autism and PD.

Postural instability is another key characteristic of PD: individuals with PD often exhibit rigidity in the neck, trunk, elbows and knees resulting in flexed posture, and struggle to readjust and maintain posture due to the loss of postural reflexes (Jankovic, 2008). Comparably, differences observed in autism include postural abnormalities in the trunk (Nobile et al., 2011, Rinehart et al., 2006a), head (Rinehart et al., 2006a) and arms (Rinehart et al., 2006b), and autistic individuals have been found to struggle to make adjustments to posture as seen in PD (Maurer and Damasio, 1982). Reduced postural stability in autism has been explicitly likened to that of PD by Minshew et al. (2004): “this type of generalized postural dysfunction [observed in autistic participants] can be seen with Parkinson disease”.

Overall, autistic movement can be likened to that of PD in core PD-diagnostic areas: bradykinesia, gait differences, and postural instability. Given that PD diagnosis relies on behavioural measures indexing these movement features, it is possible that an autistic individual would present with elevated parkinsonian traits during such an assessment.

4.2. Fine motor function

When examining performance on fine motor tasks, such as handwriting, differences have been found in both autism and PD (Beversdorf et al., 2001, Grace et al., 2017, Godde et al., 2018, Van Gemmert et al., 2003, Thomas et al., 2017, Johnson et al., 2013, McLennan et al., 1972).

Specifically, the way in which handwriting movements are made varies in autism and PD (as characterised by “kinematic features” extracted from movement trajectories). One key example is jerk – or the rate of change in the acceleration profile. Individuals in the general population move with a profile that is consistent with the minimum mean squared jerk model (Flash and Hogan, 1985, Todorov and Jordan, 1998). Here, point-to-point movements (movements from one point in space to another) are made by accelerating and decelerating gradually, which minimises jerk. By contrast, kinematic analyses have revealed increased jerk in the movement profiles of autistic individuals (Cook et al., 2013, Edey et al., 2016) and individuals with PD (Alberts et al., 2000). A lack of movement smoothness is also reflected in an increased number of sub-movements (or alternations between acceleration and deceleration) in both autism (Cook et al., 2023, Johnson et al., 2013) and PD (Lange et al., 2006, Flash et al., 1992, Castiello et al., 2000). However, distinctions may be made between autism and PD with regards to movement speed during fine motor control: whilst the slowness of movement observed in gross motor function tests in PD appears to extend to fine motor function (Alberts et al., 2000, Jankovic, 2008, Viviani et al., 2009, Lange et al., 2006, Broderick et al., 2009, Flash et al., 1992, Van Gemmert et al., 2003), fast movement profiles have been observed in autistic handwriting (Cook et al., 2013, Grace et al., 2017, Johnson et al., 2013). Thus, it is likely that autism and PD have both similarities and differences in the kinematic features of their movement profiles. It is particularly important to quantify handwriting similarities between these groups given the suggestion from an increasing number of studies to use handwriting analysis as a method for PD diagnosis (Lamba et al., 2021, Netšunajev et al., 2021, Drotár et al., 2016, Dehghanpur Deharab and Ghaderyan, 2022, Al-Yousef et al., 2020, Kamble et al., 2021, Gerger and Gümüşçü, 2022, Carvajal-Castano et al., 2022, Rios-Urrego et al., 2019). If similarities between autistic and parkinsonian handwriting features are found in direct comparison studies, the use of such methods to diagnose PD may lead to an increased misdiagnosis of PD in the autistic population.

5. Do individuals with Parkinson's disease have autistic traits?

Interestingly, a number of traits typically associated with autism have been reported in PD. Individuals with PD experience changes in cognition and social interaction as their condition develops and, as such, clinicians may be looking out for these traits. Given that many of these traits are core features of autism, it follows that autistic individuals being assessed for PD may demonstrate (socio-) cognitive differences that clinicians are aware of in the context of movement disorders.

Both autism and PD have been associated with differences in socio-cognitive processes. Differences in facial expression production have been noted in both conditions including reduced frequency and intensity (Jacobs et al., 1995, Ricciardi et al., 2015, Peron et al., 2012, Bandini

et al., 2017, Keating and Cook, 2020, Trevisan et al., 2018, Czapinski and Bryson, 2003, Loveland et al., 1994, Keating et al., 2023). Facial expressions in autism and PD are also less accurately perceived by the general population (Brewer et al., 2016, Ricciardi et al., 2017). Differences extend to the production of emotional speech (Schroder et al., 2010, Peron et al., 2012, Fusaroli et al., 2017, Hubbard and Trauner, 2007). Interestingly, a reduction in facial expressivity and prosody of speech are indexed in both PD diagnostic tools (e.g., the UPDRS) and autism diagnostic tools (e.g., the Autism Diagnostic Interview and the Autism Diagnostic Observation Schedule (Lord et al., 2000, Lord et al., 1994)). This highlights a clear link between diagnostic features of autism and PD. Alongside differences in emotion expression, a difficulty in perceiving emotional expressions produced by members of the general population is common in both conditions (Gray and Tickle-Degnen, 2010, Lozier et al., 2014, Uljarevic and Hamilton, 2013), possibly due to a mismatch in how expressions are produced by each group (Keating and Cook, 2020). Theory of mind differences are also common in both autism and PD, including less accurate situation judgement and a reduction in the use of mental state words to describe situations, with evidence from many paradigms including those using static images (Tin et al., 2018, Bodden et al., 2010), stories (Kawamura and Koyama, 2007, Peron et al., 2009, Del Prete et al., 2020, Roca et al., 2010, Tin et al., 2018) and dynamic movement-based stimuli (Wilson, 2021, Abell et al., 2000, Castelli et al., 2002, White et al., 2011, Livingston et al., 2021).

As well as socio-cognitive differences, both autism and PD have been associated with executive function differences including cognitive rigidity and a difficulty switching between tasks. Executive dysfunction is considered a core cognitive feature of PD and an impairment in task switching specifically has been observed using measures such as the Wisconsin Card Sorting Test (Dirnberger and Jahanshahi, 2013). Similarly, recent reviews evidence a lack of flexibility and difficulties in set-shifting in executive function assessments in autism (Craig et al., 2016, Westwood et al., 2016). Thus, it is likely that both groups would perform similarly on such executive function tasks. Interestingly, whilst repetitive behaviours form a core characteristic of autism (American Psychiatric Association, 2013), they are prevalent in both autism and PD (Hollander et al., 2009). In autism, a reduction in spontaneous task-switching has been associated with the severity of these behaviours (Watanabe et al., 2019). Overall, evidence supports the existence of differences in cognitive rigidity and task switching in both autism and PD.

6. Outstanding questions

6.1. What are the behavioural similarities between autism and Parkinson's disease in a matched-groups design?

This Mini-Review notes potential similarities between autism and PD in terms of gross motor function (e.g., postural instability, gait), fine motor function (e.g., handwriting), emotion expression and recognition, theory of mind ability and cognitive rigidity. However, evidence comes from distinct research silos, meaning there are many forms of variation in participant samples and experimental design which may impair the ability to directly compare results. These differences between groups may either obscure behavioural or cognitive similarities that exist, or create apparent similarities which would not exist for matched groups. One major limitation with the present research is that the older autistic adult literature is sparse, making up only 0.4 % of autism studies conducted between 2012 and 2022 (Mason et al., 2022). Indeed, only two studies indexing movement in older autistic adults were identified in a recent meta-analysis by Mason and colleagues (Mason et al., 2022). By contrast, due to the late onset of PD, studies of PD tend to recruit individuals over the age of 50. This is problematic because both movement and social cognition have been found to vary with age (Pardini and Nichelli, 2009, Ketcham et al., 2002, Maylor et al., 2002). A meaningful comparison between autism and PD must therefore be made between

similarly aged groups. Beyond age, other demographic factors that are likely to differ between samples have been associated with movement and social cognition, including gender (Kirkland et al., 2013, Baron-Cohen et al., 2015, Miller and Cronin-Golomb, 2010) and IQ (Buitelaar et al., 1999, Forti et al., 2011). This means that differences between groups may arise from variability in demographic factors. Finally, research studies employ a range of methods to index movement, emotion perception, theory of mind and cognitive rigidity, and it is unclear whether such indices are comparable. To accurately quantify similarities and differences between autism and PD, future studies must directly compare performance on the same experimental measures in a matched-groups design. Once a greater understanding of any similarities is obtained, this information should be disseminated to medical professionals to ensure an awareness of the potential overlap in symptomatology between autism and PD.

6.2. What biological factors contribute to similarities between autism and Parkinson's disease?

Whilst extant literature highlights behavioural and cognitive similarities between autism and PD, studies have not yet shed light on *why* such similarities might occur. The genetic investigations set out in the article do not provide strong evidence for a genetic relationship between autism and PD, but this does not mean a genetic relationship does not exist. Here, we reflected on evidence for a genetic relationship between autism and PD; future work may benefit from extending the investigation to other forms of parkinsonism, as well as considering relationships with specific motor phenotypes in autism as opposed to the condition as a whole (which may show greater overlap with genetic architecture related to PD motor symptoms). In addition, genetic overlap may become apparent as larger and more reliable GWAS are developed for autism and PD, and investigations using broader functional and network-based approaches may reveal overlapping systems. Finally, future genetic comparisons should consider less commonly studied forms of genetic variation such as transposable elements, which are implicated in both autism (Borges-Monroy et al., 2021) and PD (Pfaß et al., 2021, Billingsley et al., 2019b).

Irrespective of whether autistic individuals are genetically more likely to develop PD, similar presentations in autism and PD may arise if the two conditions have similar underlying biological mechanisms. This may explain the increased prevalence of PD diagnosis in the autistic population: it may not be the case that autistic individuals are indeed more likely to have co-occurring PD, rather a shared underlying biological mechanism may lead to similar presentations and thus increase the likelihood of meeting diagnostic criteria for both conditions. To validate this theory, autism and PD must share a biological mechanism which relates to motor function, as this is a core component of PD diagnosis. Dopamine is a logical candidate given its strong links to movement, having been implicated in movement speed (Bartholomew et al., 2016, Lange et al., 2006, Tucha et al., 2006), the invigoration of movements (Bova et al., 2020, Da Silva et al., 2018), generating a range of different movement speeds (Baraduc et al., 2013), modulation movement speed based on curvature (Hickman et al., 2024), and signalling the start and end points of sub-movements (Collins et al., 2016). As noted previously, dopamine dysfunction has been implicated in both PD (Rizek et al., 2016) and autism (Pavál, 2017). Future work must focus on delineating the direction of dopamine dysfunction in older autistic adults (either hyper- or hypo-dopaminergia) with respect to movement differences. In addition, directly comparing the pathology of autism and PD via in-depth neuroscientific investigations (e.g., DAT-SPECT and PET) would shed light upon overlapping biological mechanisms.

6.3. Do antipsychotic medications contribute to the prevalence of parkinsonism in the autistic population?

Antipsychotic medications are commonly prescribed to autistic individuals (Houghton et al., 2017) and have been linked to drug-induced parkinsonism (Brigo et al., 2014). Therefore, this is a critical factor to consider when interpreting elevated rates of parkinsonism in autism. Whilst the parkinsonism studies reviewed above found evidence of parkinsonism in autistic individuals who were *not* taking antipsychotic medications (Starkstein et al., 2015, Mostert-Kerckhoffs et al., 2020, Geurts et al., 2022) – suggesting that medication use alone cannot account for these findings – not all studies included a detailed analysis of medication history. For example, the large-scale prevalence studies did not systematically control for or report on antipsychotic use (Croen et al., 2015, Hand et al., 2020). Given the high prevalence of such medications in the autistic population, future research must incorporate comprehensive drug histories when assessing movement differences in autism. This will help elucidate the underlying mechanisms of such differences, be it neurodevelopmental, neurodegenerative or medication-induced.

6.4. What other factors could explain similarities between autism and Parkinson's disease?

A limitation of the current article is the sole focus on the relationship between autism and PD – this overlooks the potential involvement of other conditions. For example, elevated traits of depression, anxiety and alexithymia have been observed in both autism (Hollocks et al., 2019, Kinnaird et al., 2019) and PD (Reijnders et al., 2008, Broen et al., 2016, Alvarado-Bolanos et al., 2020). Given that these clinical traits have been associated with movement and social cognition (Nestor et al., 2022, Sachdev and Aniss, 1994, Hezel, McNally, 2014, Pijpers et al., 2005, Pisani et al., 2021, Hickman, 2019), it is possible that socio-cognitive and movement similarities between autism and PD may be the result of co-occurring clinical traits rather than the conditions themselves. Indeed, Wallace et al. (2025) reported elevated traits of depression in autistic individuals who screened positive for parkinsonism. In addition, it may be the case that autistic individuals are more likely to have co-occurring movement differences that are mistaken for PD. For example, Developmental Coordination Disorder (DCD) is a condition characterised by movement differences which has a strong co-occurrence with autism (Bhat, 2020). Future studies should quantify the involvement of co-occurring conditions in the relationship between autism and PD. Finally, an alternative explanation for increased PD diagnosis in autism is that autistic individuals are likely to have more contact with healthcare providers compared to non-autistic individuals and, consequently, they may have an increased chance of PD symptoms being detected. In practice, it is likely that increased PD diagnosis prevalence in autism is the product of many factors including similarities between conditions at behavioural, cognitive and biological/neural levels, as well as genetic contributions and healthcare factors.

7. Conclusion

A range of studies demonstrate alarmingly high rates of PD diagnosis and parkinsonism in the autistic population. However, due to a lack of research focusing on older autistic adults, it is unclear why this is the case. Whilst a number of PD risk genes have been associated with autism, current research has not yet evidenced a strong increased genetic risk of PD in autism. Interestingly, there is both anecdotal and experimental evidence that similar movement traits exist in autism and PD. It is therefore possible that autistic individuals will score highly on behavioural diagnostic tests of PD due to apparent movement similarities. This Mini-Review set out similarities between autism and PD across a range of domains, including movement (core to PD diagnostic processes) and (social) cognition (core to autism diagnostic processes).

However, direct comparison studies are needed to verify these conclusions drawn from separate research studies. Identifying the cause of the increased PD diagnosis prevalence in autism is important for the adaptation of diagnosis procedures. If autistic individuals are indeed more likely to develop PD, all stakeholders should be aware of this to ensure a prompt diagnosis. However, if autistic individuals present parkinsonian symptoms due to their autistic traits alone, clinicians should incorporate this knowledge into diagnostic assessments to prevent misdiagnosis.

Funding statement

LJH was supported by a BBSRC PhD studentship provided by the BBSRC Midlands Integrative Biosciences Training Partnership [grant reference: BB/M01116X/1]. JLC was supported by the European Union's Horizon 2020 Research and Innovation Programme under ERC-2017-StG Grant Agreement No. 757583 (Brain2Bee; Jennifer Cook PI).

Authorship contribution statement

LJH and JLC conceptualised the article. LJH conducted the review of the literature and wrote the manuscript. LJH and JLC edited the manuscript.

Acknowledgements

We thank our research participants for allowing us to include their quotes in this article.

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